

Reevaluating Advanced NFL Metrics

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Executive Summary

In the highly competitive ecosystem of the National Football League (NFL), front offices rely on metrics like Success Rate and Explosive Plays to evaluate offensive efficiency. Historically, these metrics rely on decades-old thresholds: gaining 40%, 60%, and 100% of required yardage across early downs to qualify as a success, and defining explosive plays as 10-yard rushes or 20-yard passes. Because the sport's evolution may have rendered these benchmarks inaccurate, this study evaluated regular-season plays from the past 5 years of the *nflverse* dataset. To isolate genuine competitive execution, the data was filtered to meaningful plays. By analyzing yardage gained against Expected Points Added (EPA) and subsequent scoring probabilities, the analysis identified the optimal crossover points where a play yields a net-positive expected outcome.

The evaluation revealed that legacy benchmarks are overly lenient. For Success Rate, the EPA analysis established that a stricter framework—requiring 50% of yardage on first down, 70% on second, 90% on third, and 100% on fourth—are better benchmarks to decide if an offense is staying on schedule. This calibration reduced the standard deviation of success across early downs.

Furthermore, the data showed no justification for applying separate thresholds for runs versus passes in evaluating explosive plays. By balancing predictive power with play availability, the study identified a play-agnostic 15-yard gain as the optimal Explosive Play benchmark, which outperformed traditional metrics at tight game margins by filtering out the noise of low-impact rushes. To validate the framework, an Ordinary Least Squares regression model evaluated season-level Points Per Game. The revised metrics demonstrated improved explanatory power, yielding a higher R-squared (0.6959) and lower Akaike Information Criterion (729.83) than legacy models. This demonstrates that these revised benchmarks provide a more accurate signal for forecasting NFL outcomes.

Problem Statement and Business Context

The National Football League (NFL) operates as a multi-billion dollar enterprise where tiny advantages decide on-field outcomes. The availability of comprehensive play-by-play data has forced front offices to transition from qualitative scouting toward quantitative analysis. Finding stable, predictive metrics translates to improved salary cap allocation, schematic advantages, and team revenue. Meanwhile, the explosive growth of the sports betting market has created a secondary financial ecosystem where improved data and metrics have the potential for immense monetary value.

Advanced metrics have been created to supplement traditional metrics such as yards per play. Two such advanced metrics are Success Rate and Explosive Plays. Historically, offensive

efficiency was measured by raw "Yards Per Play." However, this metric treats all yards equally, failing to account for the mathematical leverage of down and distance. Success Rate solves this by measuring a team's ability to stay "on schedule" and ensure manageable third downs. Success Rate has proven to be a more stable forecasting tool over the course of a season than raw yardage. The accepted legacy standard demands that an offense acquire 40% of required yardage on first down, 60% on second down, and 100% on third or fourth down. Explosive Plays, another advanced metric, measure chunk yardage, historically defined as a rush of 10 or more yards or a completion of 20 or more yards.

This analysis explores whether these thresholds are the most stable and predictive benchmarks for these metrics. It will also determine if revised benchmarks can better capture offensive success.

Data Description and EDA Findings

The primary dataset for this analysis is derived from the nflverse project, a comprehensive, open-source repository containing NFL play-by-play and scheduling data. This dataset includes 1,279,628 individual play records across 27 seasons (1999-2025) and 372 distinct variables. Data acquisition was executed within a Python environment utilizing the nflreadpy library, allowing for the direct download of game logs into polars and pandas DataFrames.

Methodology

Preprocessing:

Play-by-play data from the nflverse repository was downloaded for all available seasons (1999–2025). To isolate the analysis to meaningful plays, the scope of the dataset was restricted to regular-season matchups and focused exclusively on traditional passes and rushes from scrimmage. Non-standard events - such as two-point conversion attempts, clock-management plays (quarterback kneels and spikes), and snaps distorted by accepted penalties - were removed. Strict filters were also applied to discard incomplete records missing variables like down, possessing team, Expected Points Added (EPA), and yards gained. Finally, to eliminate statistical noise, a competitive game-state filter was enforced: only plays where the possessing team's pre-snap win probability rested between 10% and 90% were included. This ensures the resulting metrics reflect authentic competitive execution rather than conservative clock-draining strategies or soft prevent defenses.

Feature Engineering:

The following features were created and added to the dataframe:

conversion_pct_gained: Converts raw yards gained into a proportion of the yards required for a first down (yards gained / yards-to-go).

pct_gained_bin: bins the conversion percentage strictly between 0.0 and 1.0, rounding down to the nearest 10% interval (e.g., 0.4, 0.5) to group plays into uniform buckets.

is_pos_epa: Creates binary flags for plays that generate positive Expected Points Added.

is_explosive: Creates binary flags for plays that meet a minimum yardage threshold.

is_success_benchmark: Flags plays as "successful" using the legacy baseline rules: gaining at least 40% of needed yards on 1st down, 60% on 2nd down, and 100% on 3rd/4th down.

revised_success_benchmark: Evaluates success using your newly calibrated thresholds: gaining at least 50% on 1st down, 70% on 2nd down, 90% on 3rd down, and 100% on 4th down.

Model Selection:

This analysis relies primarily on descriptive statistics to establish new offensive benchmarks. To validate these findings, an Ordinary Least Squares multiple linear regression model was selected. By regressing Points Per Game against both the traditional and revised sets of Success Rate and Explosive Play features, the analysis determines which metrics possess greater explanatory power. This approach tests whether the revised benchmarks provide a stronger, more accurate signal for offensive success.

Results and Evaluation

Success Rate by Down

To determine the optimal thresholds for offensive success, we evaluated the relationship between yardage gained and EPA. Figure 1 illustrates this dynamic for the modern NFL era (2021–2025). By plotting the proportion of Yards-to-Go gained against the resulting Average EPA, the chart shows the metric at which a play transitions from a net negative outcome to a net positive one for the offense. This was segmented by down.

The revised benchmark indicators (represented by the star markers) mark the minimum yardage threshold required for an average EPA that is positive (downs in bold represent a different benchmark than the traditional thresholds):

- **1st Down: 50%**
- 2nd Down: 60%
- **3rd Down: 90%**
- 4th Down: 100%

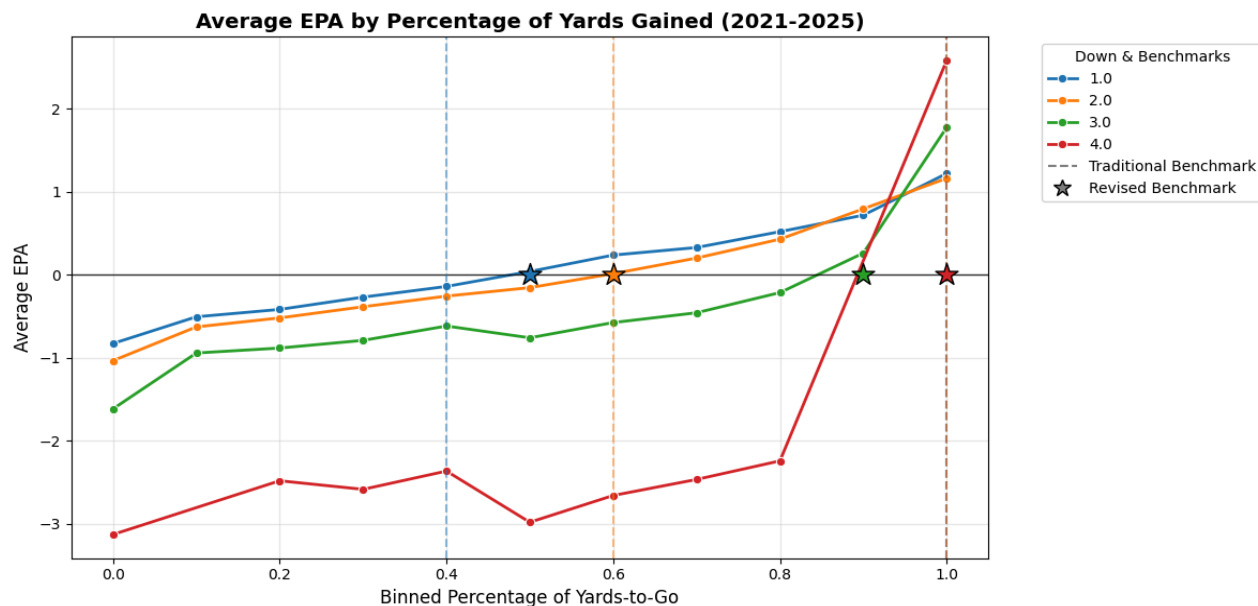


Figure 1: Average EPA by Binned Percentage of Yards-to-Go (2021-2025, aggregated)

Extending this same analysis for each individual season in the dataset reveals consistent metrics for 1st, 3rd and 4th down, as shown in Figure 2. Meanwhile, the threshold for 2nd down oscillates between 60% and 70%.

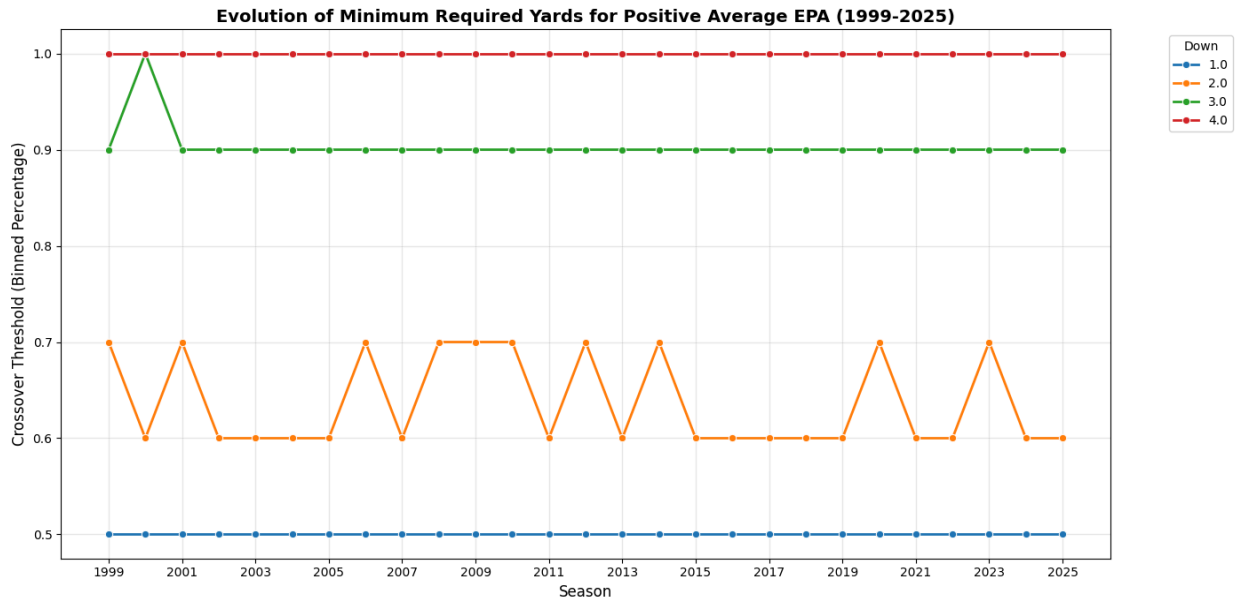


Figure 2: Average EPA by Binned Percentage of Yards-to-Go (1999-2025)

Because averages can be heavily influenced by outliers, another analysis was conducted using positive EPA rate. Now the optimal threshold is determined by the minimum threshold in which a majority of its plays contain a positive EPA, as shown in Figure 3. Applying this analysis method

to the most recent 5 years of data, the optimal threshold for 2nd down is revealed to be 70%, while the thresholds for all other downs remain constant:

- **1st Down: 50%**
- **2nd Down: 70%**
- **3rd Down: 90%**
- **4th Down: 100%**

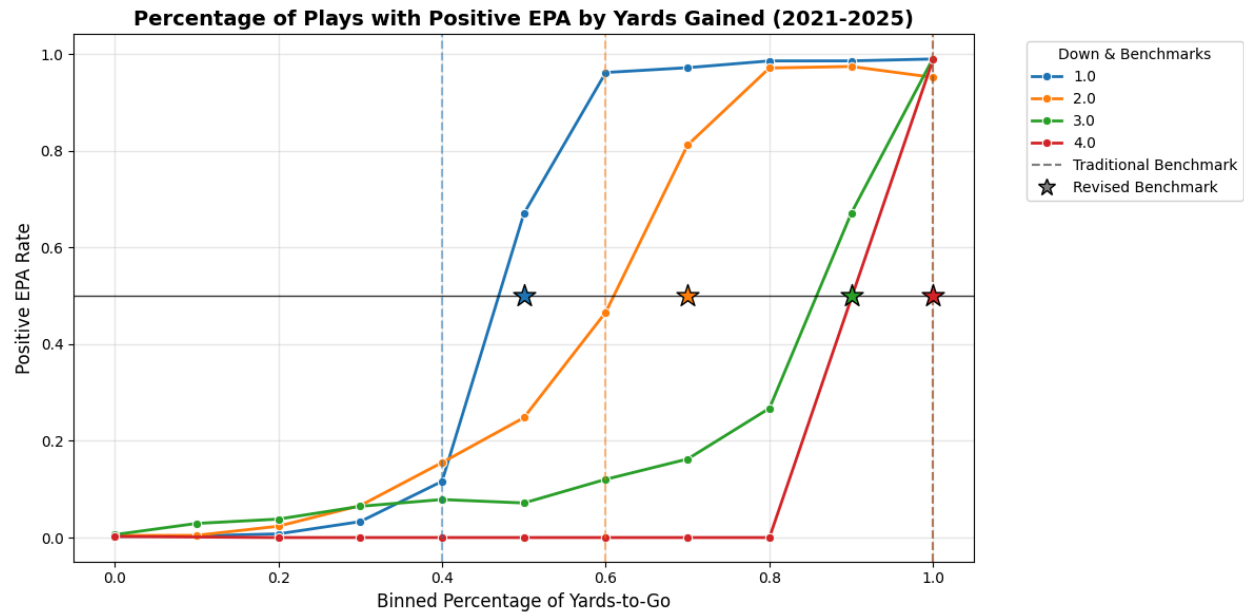


Figure 3: Positive EPA Rate by Binned Percentage of Yards-to-Go (2021-2025)

Figure 4 shows that once again extending this analysis to each individual season included in our dataset produces a very similar chart to Figure 2. Once again, thresholds remain constant across all downs but 2nd down, which again oscillates between 60% and 70%.

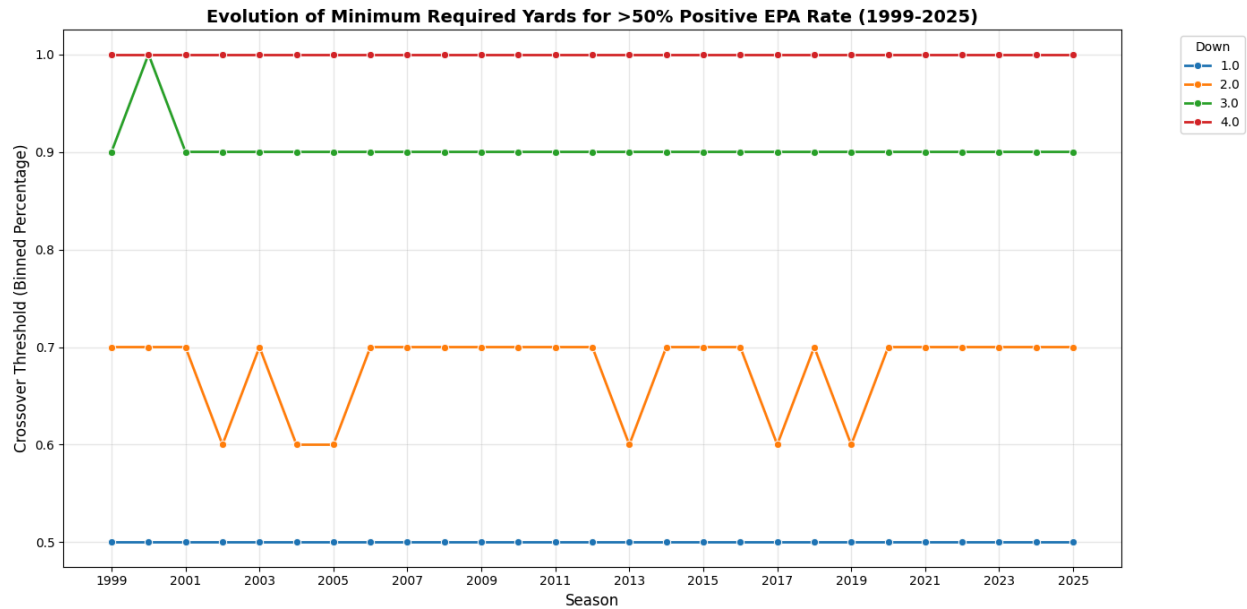


Figure 4: Positive EPA Rate by Binned Percentage of Yards-to-Go (1999-2025)

These analyses suggest that the threshold for determining success rate should be raised for early downs - from 40% -> 50% for 1st down, and possibly from 60% -> 70% for 2nd down (though the data suggest that either 60% or 70% would make a suitable benchmark). Interestingly, the analysis suggests lowering the threshold for 3rd down from 100% to 90%. This posits that gaining 90% of the yards-to-go on 3rd down is enough to keep the offense “on schedule” and should be considered a success. For such cases to truly be considered a success, however, the offense would need to attempt the 4th down conversion on the subsequent play. NFL data analysts have long contended that NFL offenses should attempt 4th down conversions at higher rates than have historically been observed.

To see if these revised metrics represent an improvement on the Success Rate definition, we calculated the success rate by down for both metrics, as shown in Figures 5 and 6.

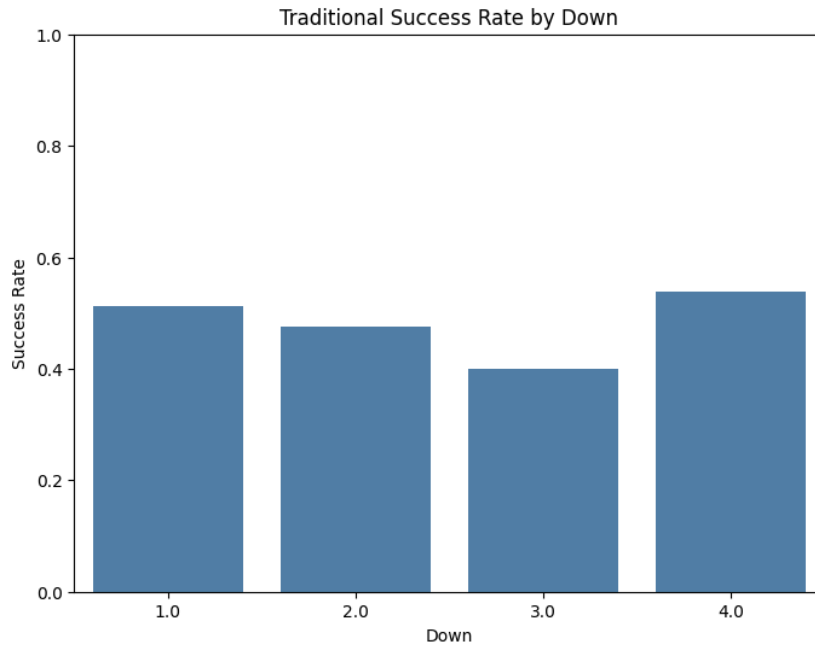


Figure 5: Success Rate by Down using Traditional Thresholds

The standard deviation for success percentage for 1st-3rd down is 0.0555 for the traditional metric. Note that 4th down is not included in this calculation - because the consequence of an unsuccessful 4th down attempt results in a turnover, 4th down will retain a higher success rate due to selection bias resulting from only attempting conversions when success is likely.

Overall, comparing the traditional success rate to successful plays as defined by having a positive EPA for each individual play reveals the following metrics:

--- Confusion Matrix ---

True Positives (Correctly labeled Success): 53869
False Positives (Labeled Success but negative EPA): 7527
True Negatives (Correctly labeled Failure): 64070
False Negatives (Labeled Failure but positive EPA): 2040

--- Classification Metrics ---

Accuracy: 0.9250
Precision: 0.8774
Recall: 0.9635
F1 Score: 0.9184

Overall, these two metrics agree on 92.5% of plays. A relatively high recall of .9635 and low precision of 0.8774 between these two metrics suggest that the traditional thresholds are too inclusive and need to be more selective.

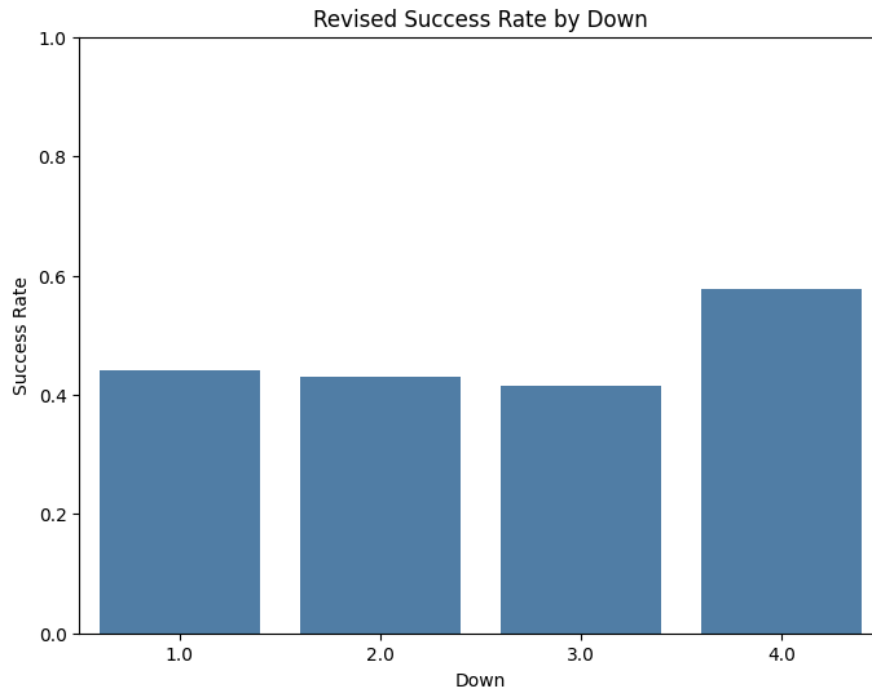


Figure 6: Success Rate by Down using Revised Thresholds

Standard deviation for success percentage for 1st-3rd down is 0.0136 for the revised metric. This demonstrates a tremendous drop from 0.0555 for the traditional metric, providing evidence that the revised metrics represent a much more consistent success rate across the first 3 downs. The complete confusion matrix is shown below:

```
--- Confusion Matrix ---
True Positives (Correctly labeled Success): 52567
False Positives (Labeled Success but negative EPA): 2802
True Negatives (Correctly labeled Failure): 68795
False Negatives (Labeled Failure but positive EPA): 3342

--- Classification Metrics ---
Accuracy: 0.9518
Precision: 0.9494
Recall: 0.9402
F1 Score: 0.9448
```

The overall agreement rate for the revised metrics rose to 95.18%. Unlike with the traditional benchmarks, precision (0.9494) and recall (0.9402) are similar in value. This suggests that the more selective criteria for success rate used in the revised metrics produce a better calibrated model.

Explosive Plays

Pass plays are much more likely to result in large gains, as shown in Figure 7. The traditional threshold for Explosive Plays addresses this by bifurcating the yardage needed to be defined as an explosive play based on whether the play is a pass or run. Thresholds vary among organizations - from 20 yards for a passing play and 10 yards for a rushing play to 16 and 12 yards, respectively.

To justify a lower threshold needed for a running play to be classified in the same category as a passing play, one would expect a higher average EPA for 10-12 yard runs than passes of similar yards. But the data reveal that at this yardage, EPA values are closely aligned for runs and passes, as shown in Figure 8. In fact, pass plays actually have slightly higher EPA values.

This suggests that the methodology of using separate thresholds for runs and passes to define an explosive play is unsound. If the goal of the explosive play metric is to identify a yardage benchmark for plays that add significantly to a team's scoring chance, there is no justification for treating run and pass plays differently. Going forward, this analysis will seek to identify the best yardage benchmark to define explosive plays, regardless of play type.

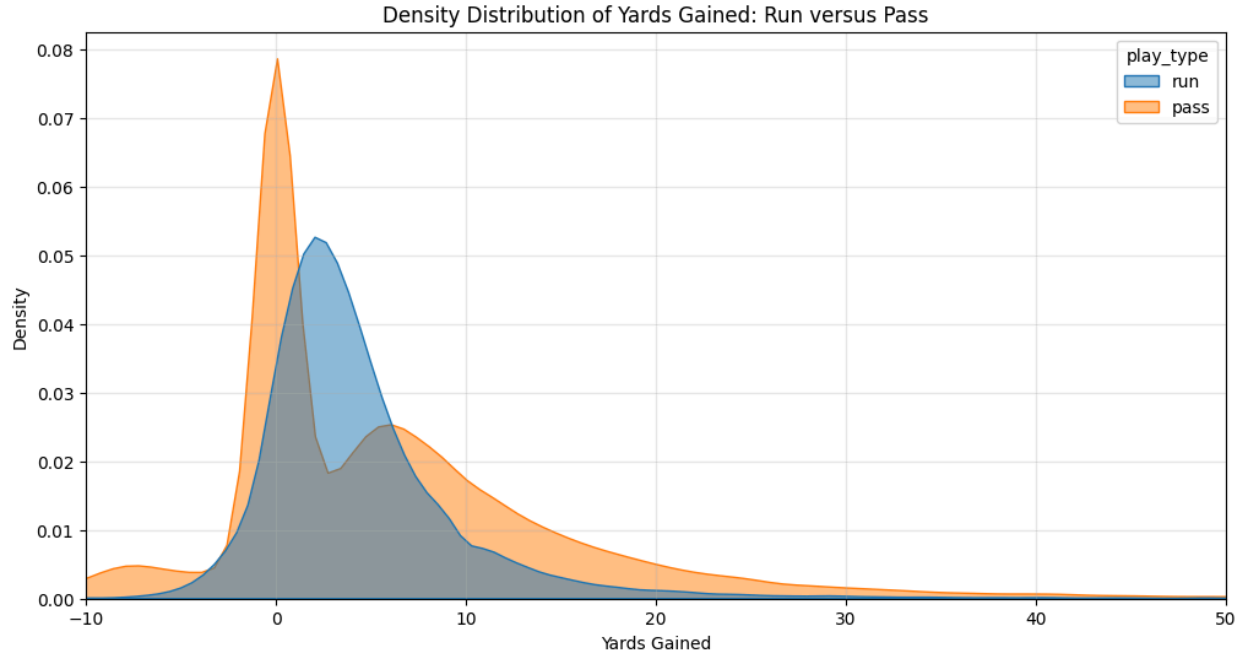


Figure 7: Density Distribution of Yards Gained: Run vs Pass

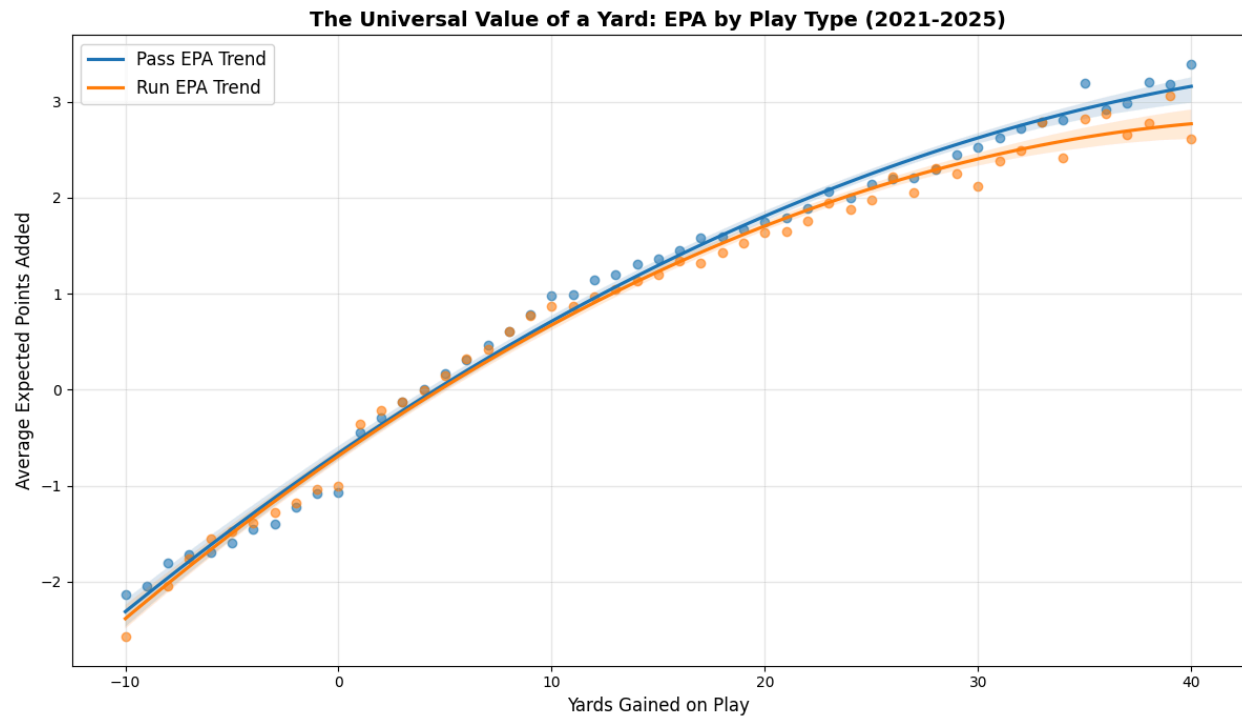


Figure 8: Average EPA by Yardage: Pass v. Run

To find this optimum threshold, this analysis looks at two factors important in determining the minimum benchmark for an explosive play: Predictive Power and Availability.

Predictive Power asks: for a given yardage threshold, how likely does a drive containing an Explosive Play end in a score? Plotting Scoring Probability against Yardage Threshold, the green line in Figure 9 shows a steady, smooth rise in this relationship. As the Yardage Threshold increases, so does the Scoring Probability. The greater the Yardage Threshold chosen for this metric, the more predictive it is at forecasting scoring.

But for this metric to be useful, it also has to be relatively common in NFL games. Availability asks how often plays occur in an NFL game, based on Yardage Threshold. Availability has an inverse relationship to Predictive Power, as shown in Figure 10. As the Yardage Threshold increases, Availability decreases. Figure 11, meanwhile, looks at the Explosive Play differential by Yardage Threshold, a commonly cited use of Explosive Plays.

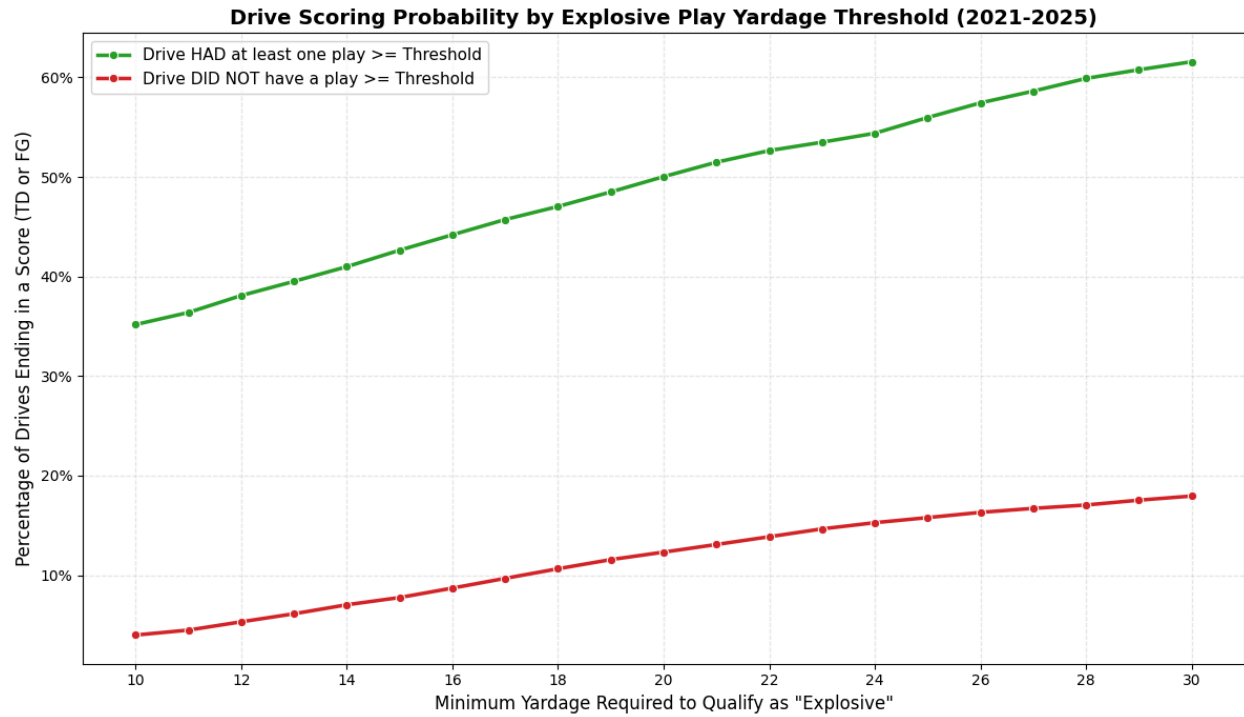


Figure 9: Scoring Probability by Explosive Play Threshold

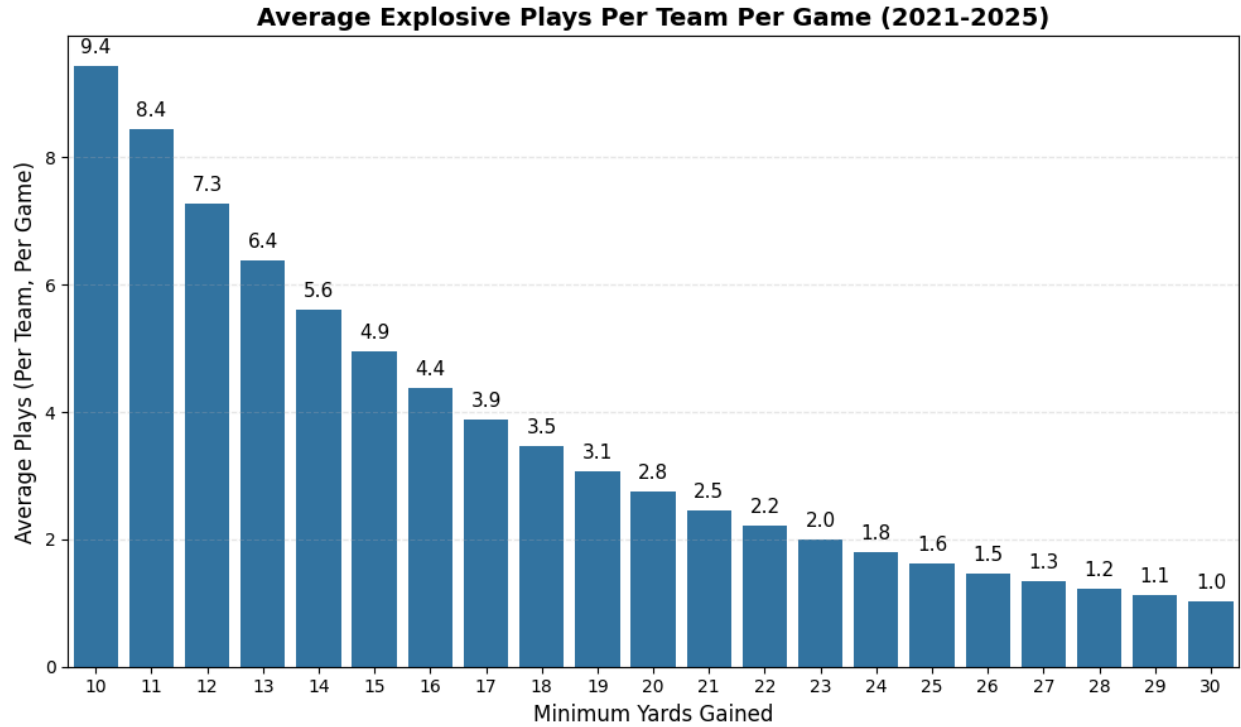


Figure 10: Availability of Explosive Plays by Yardage Threshold

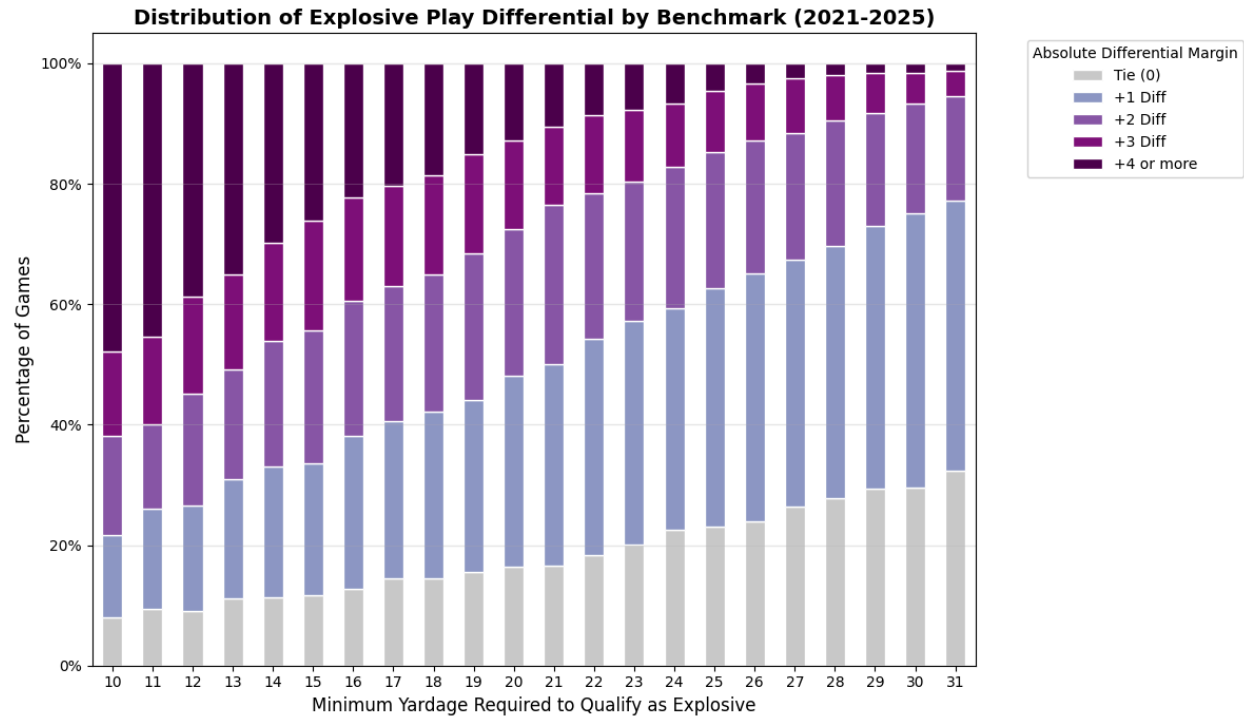


Figure 11: Availability of Explosive Play Differential by Yardage Threshold

The ideal benchmark for Explosive Plays will balance these two metrics - it will be Predictive enough to be meaningful, and Available enough to be useful. Taking these two considerations into account, we suggest that an improved threshold for designating a play as Explosive is a **play-agnostic gain of 15 yards or more**.

To evaluate if this revised metric represents an improvement over the traditional benchmarks, we plotted win percentage by explosive play differential, as shown in Figure 12. The game data backs up the revised thresholds, which outperform the original metrics at predicting win probability. This is especially true for lower values of Explosive Play Differential (64% v. 57% for a +1 differential), while the predictive value of the two metrics converge at larger differentials (82% v. 81% for a +4 or more differential).

Overall, these win probabilities demonstrate that a revised metric for Explosive Plays of 15+ yards represents an improvement on the traditional metrics of 10+ yards for runs and 20+ yards for passes.

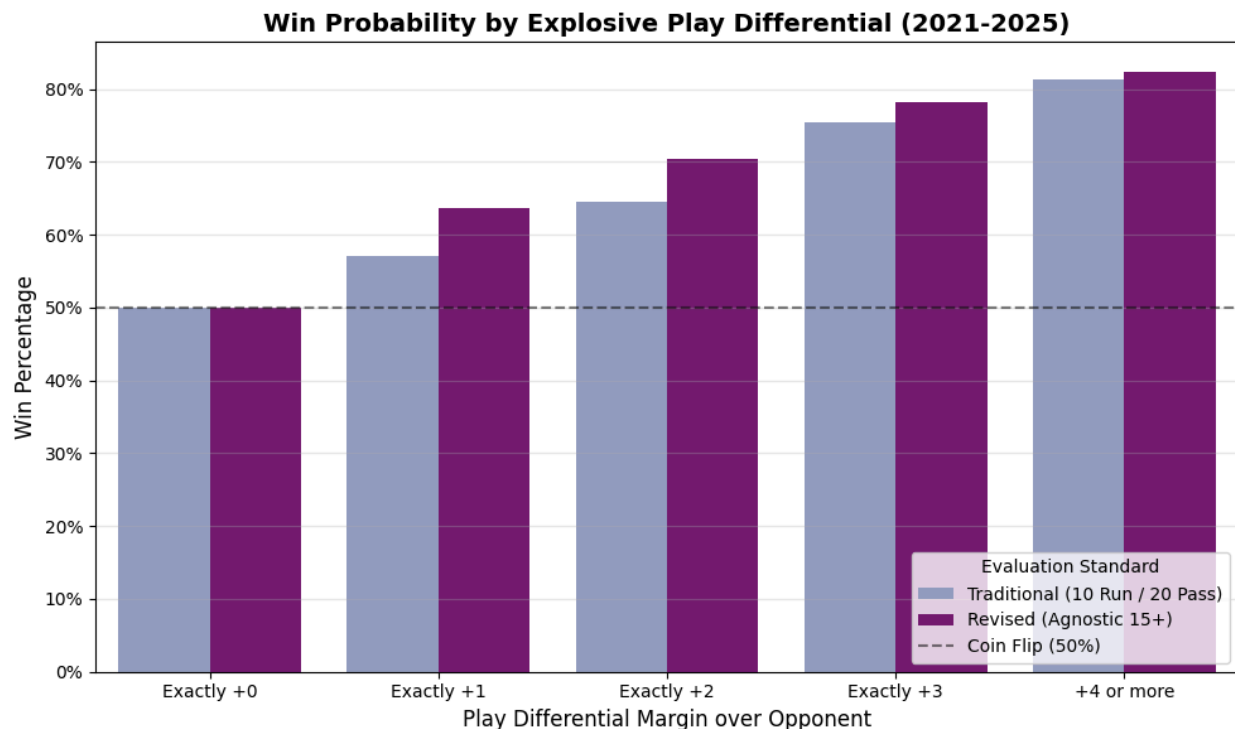


Figure 12: Win Percentage by Explosive Play Differential: Traditional v. Revised

For a final analysis, a multiple linear regression model was deployed to evaluate how well these engineered benchmarks explain variations in season-level Points Per Game (PPG). When comparing the traditional framework against the revised 50/70/90 success rate and agnostic 15-yard explosive threshold, the revised metrics consistently demonstrated stronger explanatory power. For offensive scoring, the revised model yielded a higher R-squared (**0.6959** compared to **0.6695**) and a notably lower Akaike Information Criterion, or AIC (**729.83** versus **743.11**), indicating a more efficient and accurate model fit with less information loss. A similar, though more marginal, outperformance was observed on the defensive side of the ball, where the revised framework again edged out the legacy metrics in both R-squared and AIC. The full results of this analysis are shown below:

```
==== PPG_FOR | TRADITIONAL (40/60/100 SR | Split Explosive) =====
R-squared: 0.6695
Adj. R-squared: 0.6653
AIC: 743.11
```

```
==== PPG_FOR | REVISED (50/70/90/100 SR | Agnostic 15+) =====
R-squared: 0.6959
Adj. R-squared: 0.6920
AIC: 729.83
```

```
==== PPG_AGAINST | TRADITIONAL (40/60/100 SR | Split Explosive) =====  
R-squared: 0.4322  
Adj. R-squared: 0.4249  
AIC: 725.54
```

```
=== PPG_AGAINST | REVISED (50/70/90/100 SR | Agnostic 15+) =====  
R-squared: 0.4346  
Adj. R-squared: 0.4274  
AIC: 724.85
```

Ultimately, this regression analysis provides the mathematical confirmation that the stricter, newly calibrated benchmarks capture a significantly clearer and more predictive signal for actual game scoring than the legacy definitions they replace.

Limitations and Future Work

While the revised metric framework demonstrates significant improvements in explanatory power over legacy benchmarks, this analysis is subject to certain limitations. The methodology relies exclusively on regular-season play-by-play data, omitting the heightened competitive intensity of the postseason. Furthermore, the ordinary least squares (OLS) regressions function primarily as descriptive evaluation tools rather than a true method of forecasting. Future research could transition these findings from descriptive to prescriptive by employing logistic regression or machine learning classification algorithms to forecast binary win/loss outcomes on unseen future seasons. Additionally, incorporating spatial player-tracking variables (such as Next Gen Stats) and opponent-adjusted efficiency weights could further refine these thresholds, evolving the metrics beyond pure yardage benchmarks to capture a more complete context of an NFL game.

Ethical Considerations

The ethical implications of this analysis primarily concern the application of the engineered metrics rather than data privacy. The foundational data sourced from the *nflverse* consists entirely of public, professional performance records devoid of sensitive personal information, mitigating traditional data privacy risks. However, ethical consideration should be applied to how these revised benchmarks are used by front offices. Because the 50/70/90 success rate and 15-yard explosive thresholds evaluate play-level outcomes, they are dependent on the execution of all eleven offensive players. Utilizing these team-dependent metrics to evaluate, penalize, or compensate individual players during contract negotiations, without isolating individual contribution, could lead to unfair labor practices and inequitable compensation. In addition, analysts must remain aware that historical play-by-play data by nature reflect existing biases and conservative coaching methods; deploying these models without human oversight risks reinforcing historical biases rather than objectively advancing the game.